DAY 10 | PRE-READ

Building Screens from Blocks

A light read before Day 10. About 15 minutes, no code required.

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Why read this?

For two weeks you built the back end: the kitchen of the restaurant. Today you start on the dining room, the screens people actually see and touch. You will use **React**, the most popular tool for building web app screens, and you will build the first SmartDesk dashboard with OpenAI Codex.

1. Apps that never reload

Old websites reloaded the whole page every time you clicked, like reprinting a full newspaper to fix one headline. Modern apps load once and then update only the part that changed. These are called **single page applications**, or SPAs.

2. Screens made of building blocks

Look at any app screen. It is made of repeated pieces: a header, a search bar, cards, a button. React calls each piece a **component**. You build a ticket card once and reuse it for every ticket, just like stacking the same toy block many times.

3. Inputs and memory

When the user does something (an event), the component updates its memory (state), and React redraws that part of the screen automatically.

Idea	React name	Everyday example
Information handed to a block	props	A name badge printed with your name
Information a block remembers	state	A scoreboard that changes during a match
Something the user does	event	Pressing a button or typing

Key idea: you describe what the screen should look like for the current data. React works out how to update it.

4. About today's assessment

Today includes Basics assessment 2, with multiple choice and coding questions on SQLAlchemy, JavaScript, Node.js and React basics. Re-read your post-reads for Days 6, 8 and 9 and re-run one notebook.

5. Think about these before class

- Open your favourite app and list five components you can see on one screen.
- Which parts of that screen change when you tap something, and which stay the same?
- If Codex wrote a screen for you, how would you check it is correct and safe?

Tsumiki (積み木): building blocks